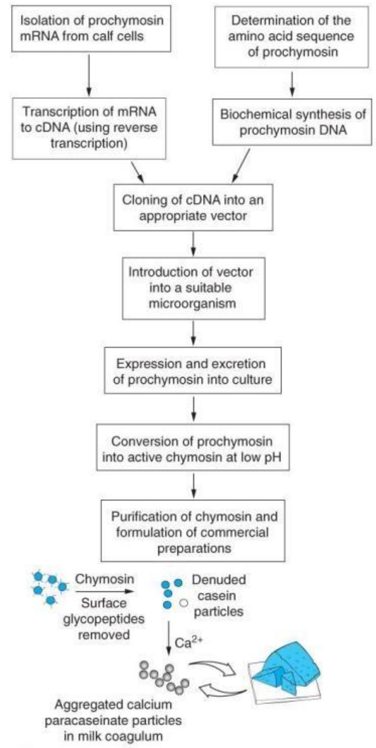
# Proteins as Biotechnology Products

• Use of proteins in manufacturing is a time-tested technology

– Beer brewing and winemaking

– Cheese making

The use of protein to make product was used from along time to make cheese or to make wine, beer but the product was limited, after the discovery of recombinant DNA the number of product increased, it allow also the produce of specific protein with specific function




• Recombinant DNA technology made it possible to produce specific proteins on demand

– Enzymes

– Hormones

– Antibodies

Now as we said before all of these things weren't produced before but after the discovery of this technology, we produced it this help us in many things including making drug for human or other species

**TABLE 4.1 SOME ENZYMES AND THEIR INDUSTRIAL APPLICATIONS**

| Enzyme | Application |
|---|---|
| Amylases | Digest starch in fermentation and processing |
| Proteases | Digest proteins for detergents, meat/leather, cheese, brewing/baking, animal/human digestive aids |
| Lipases | Digest lipids (fats) in dairy and vegetable oil products |
| Pectinases | Digest enzymes in fruit juice/pulp |
| Lactases | Digest milk sugar |
| Glucose isomerase | Produce high-fructose syrups |
| Cellulases/hemicellulases | Produce animal feeds, fruit juices, brewing converters |
| Penicillin acylase | Produces penicillin |

## Biotech Drugs and other medical applications

– Produced through microbial fermentation or mammalian cell culture mammalian cell culture: is the growth of animal cell in vitro (outside the cell)

– Complicated and time-consuming process

As we know before the biotech drug involve the use organisms so its not an easy process and of course it will take time and sometime its expensive because some time we will use a whole organism to produce protein

– Once the method is determined produce large batches of the protein products in bioreactors by growing the host cells that have been transformed to contain the therapeutic gene of interest

- Cells are stimulated to produce the target proteins through precise culture conditions that include a balance of temp., oxygen, acidity, and other variables

You have to know that if we want to produce protein outside its normal cell for example if we want to produce human protein in bacteria its important to provide the same condition as in the normal cell

- At appropriate time the proteins are isolated from the cultures, tested at every step of purification and formulated into pharmaceutically active products

**TABLE 4.2 SOME PROTEIN-BASED PHARMACEUTICAL PRODUCTS (MOST PRODUCED AS RECOMBINANT PROTEINS)**

| Protein | Application |
|---|---|
| Erythropoietins | Treatment of anemia |
| Interleukins 1, 2, 3, 4 | Treatment of cancer, AIDS; radiation- or drug-induced bone marrow suppression |
| Monoclonal antibodies | Treatment of cancer, rheumatoid arthritis; used for diagnostic purposes |
| Interferons ($\alpha$, $\beta$, $\gamma$, including consensus) | Treatment of cancer, allergies, asthma, arthritis, and infectious disease |
| Colony-stimulating factors | Treatment of cancer, low blood cell count; adjuvant chemotherapy; AIDS therapy |
| Blood clotting factors | Treatment of hemophilia and related clotting disorders |
| Human growth factor | Treatment of growth deficiency in children |
| Epidermal growth factor | Treatment of wounds, skin ulcers, cancer |
| Insulin | Treatment of types 1 and 2 diabetes mellitus |
| Insulin-like growth factor | Treatment of type 1 diabetes mellitus |
| Tissue plasminogen factor | Treatment after heart attack, stroke |
| Tumor necrosis factor | Cancer treatment |
| Vaccines | Vaccination against hepatitis B, malaria, herpes |

- Applications of Proteins in Industry

– Food processing

Like cheese and wine

– Textiles and leather goods

Like the leather jacket we can use protein that work as a substitute for the real leather

– Detergents

For example, the washing powder

– Bioremediation: treating pollution with proteins and clean up harmful wastes with proteins

• Proteins

– Are complex molecules built of chains of amino acids ( we have 20 different amino acids each will differ only in side chain, 9 are essential these essential can't be made by the body as a result we have to get it form food)

– Have specific molecular weights

– Have electrical charge that causes them to interact with other molecules

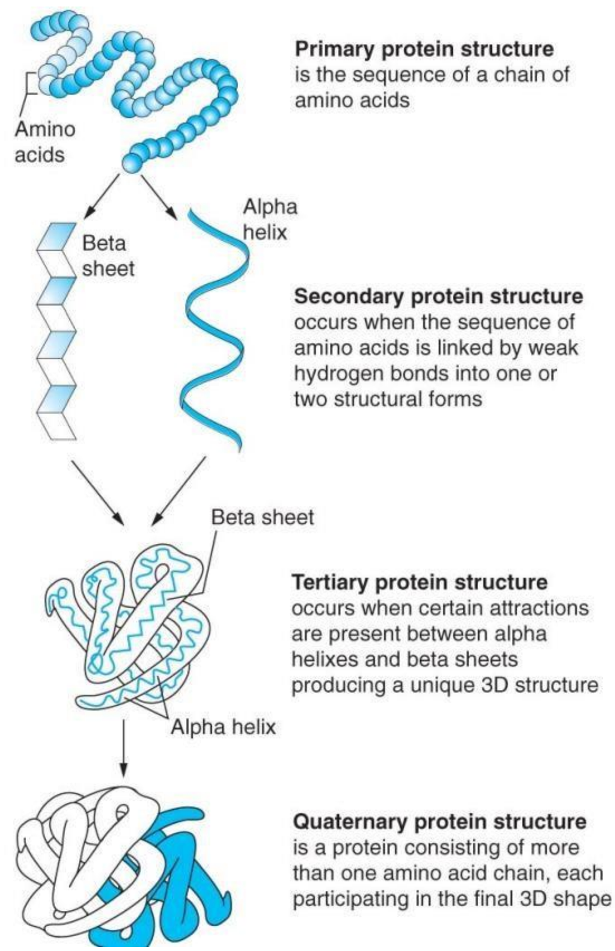
• Hydrophilic – water loving

• Hydrophobic – water hating

## Structural Arrangement – four levels

– **Primary structure** is the sequence in which amino acids are linked together

– **Secondary structure** occurs when chains of amino acids fold or twist at specific points forming new shapes due to the formation of hydrogen bonds between hydrophobic amino acids




• Most common shapes are alpha helices and beta sheets

– Alpha helix = amino acids form right handed spiral so hydrogen bonds stabilize the structure linking an amino acid's nitrogen atom to the oxygen atom of another amino acid –

Beta sheet = hydrogen bonds link nitrogen and oxygen atoms forming sheets that are parallel (chains run in same direction or anti-parallel (chains alternate in direction)

– Tertiary structures = 3 dimensional polypeptides and are formed when secondary structures are cross linked

• Tertiary structure of protein determines its function

– Quaternary structures are unique, globular, three-dimensional complexes built of several polypeptides

# Protein Folding

– The structure and function of a protein depends on protein folding

– If protein is folded incorrectly, desired function of a protein is lost, and a misfolded protein can be detrimental

– 1951: two regular structures were described

• Alpha helices and beta sheets

• Structures are fragile; hydrogen bonds are easily broken

• Incorrectly folded proteins can lead to diseases including Alzheimers, cystic fibrosis, mad cow, forms of cancer and even some heart attacks

• *** a challenge of biotechnology is to understand and control the protein folding in the manufacturing process

Because as we said before the function of protein depends on the folding so if we cant control the folding the function of protein may be lost

• Glycosylation – post-translational modification wherein carbohydrate units are added to specific locations on proteins

• More than 100 post-translational modifications occur within eukaryotic cells

• This change can affect the protein's activity by increasing:

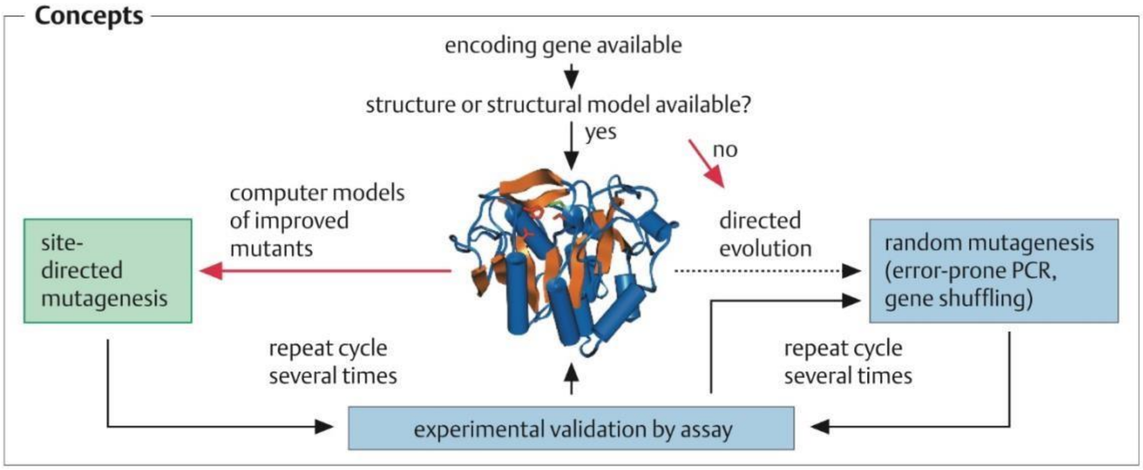
– its solubility

– orient proteins into the membranes

– extend the active life of a molecule in an organism

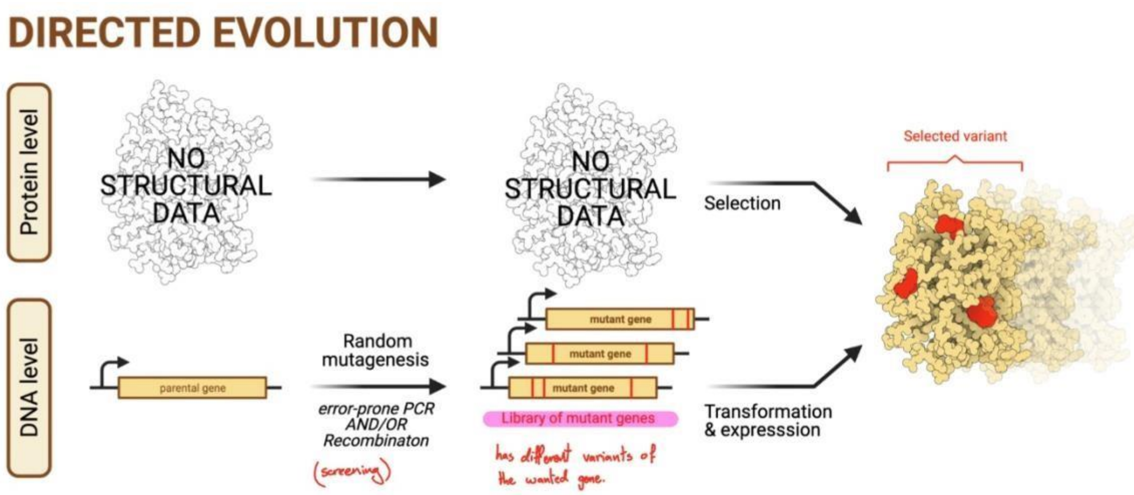
## Protein engineering

Can be done by two major concepts (depending in how much information you have of the protein):

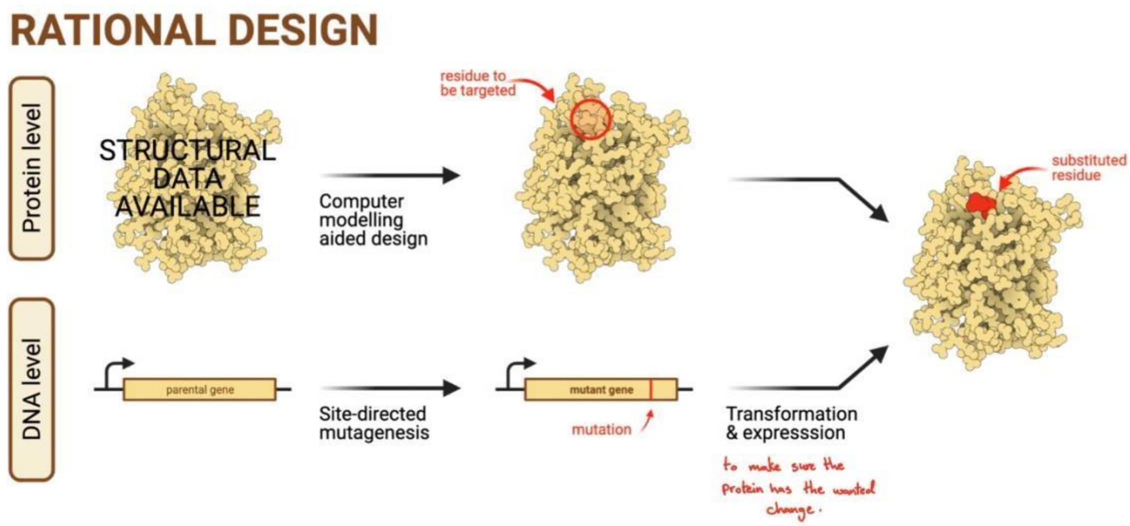



1- Site-directed mutagenesis
2- Directed evolution

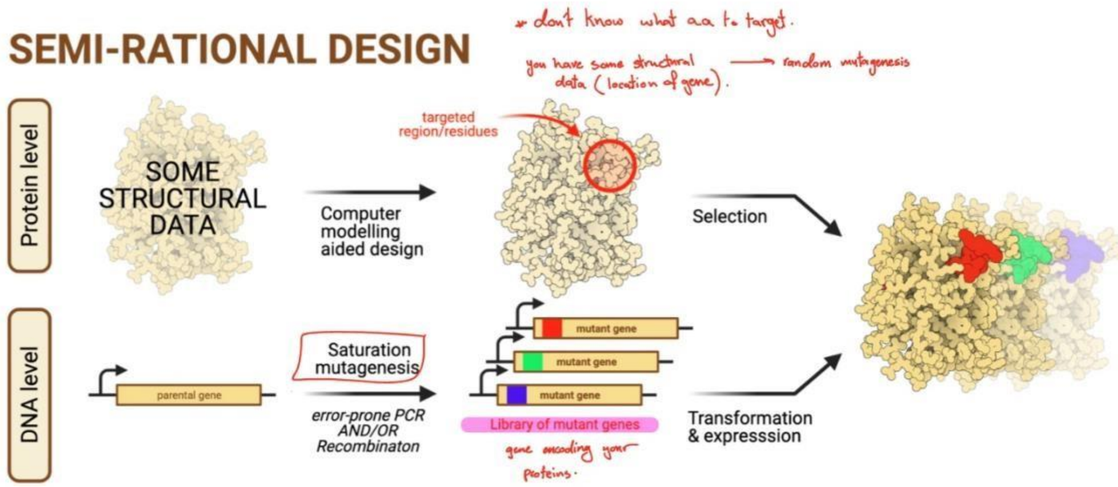
- Three protein engineering approaches:
1- Directed evolution




2- Rational design, depends on knowledge of proteins (site directed mutagenesis)




3- Semi rational design (combination of rational and directed evolution)



| | Directed evolution | Rational design | Semi-rational design |
|---|---|---|---|
| **Parental gene** | A single gene or a group of homologous sequences | A single gene | A single gene |
| ***A priori* knowledge requirement** previous knowledge | Not required | Required | Required |
| **Genetic diversity creation** type of mutagenesis | Random mutagenesis or DNA recombination | Focused mutagenesis | Focused mutagenesis |
| **Library size** | Large | Small | Small to medium |
| **Screening** | High to ultra high throughput | Low to high throughput | Low to high throughput |
| **Advantages** | • No prior knowledge of the enzyme structure and mechanism is required.<br>• Mutate the entire enzyme, and as such, it is possible to identify mutations distant to the active site that affect the enzymatic activity via allosteric interaction. | • Small library size.<br>• Less time and<br>• effort on screening.<br>• Particularly<br>• advantageous when there is no high-throughput screening system available. | • Library size is significantly reduced compared to directed evolution.<br>• A larger portion of the protein sequence space is explored compared to rational design. |
| **Disadvantages** | • Large library size.<br>• Impossible to explore the<br>• full protein sequence space, even with the most powerful selection or screening method.<br>• Time consuming to develop an assay and to screen large library.<br>• Resource intensive. | • *A priori* knowledge is required.<br>• Mutations are mainly targeted at the active site. | • *A priori* knowledge is required.<br>• Mutations are mainly targeted at the active site. |

# Mutagenesis

Mutagenesis: the process of introducing mutations.

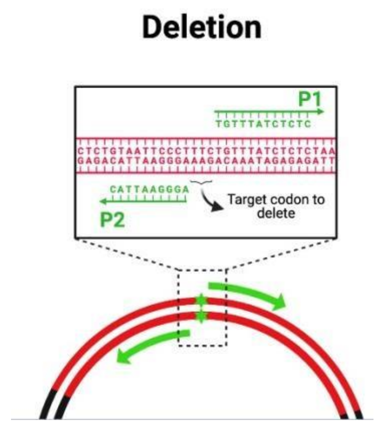
There are 3 mutagenesis approaches, to change the codons in the gene coding your protein so that change will reflect on the production of your gene:

1- Focused (site directed): inserts mutations to a specific region within a gene.
2- Random mutations: on random location, hoping to produce changes in the locations coding your proteins.
3- Recombination: exchanging large segments between homologous genes, creating **chimeras**.

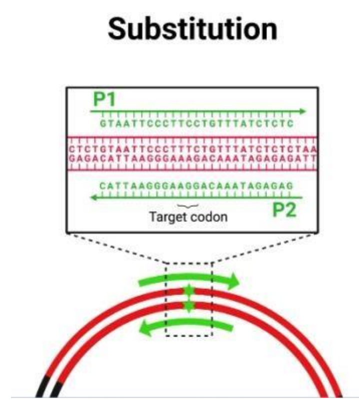
# Site directed mutagenesis

Site directed mutagenesis changes the dna sequence encoding the target protein to encode a different amino acid at the selected locations. It also facilitates the introduction of specific predefined sequence alterations into protein's backbone.

It includes:




1- **Deletion**, primers bind to areas before and after the target codon, whole sequence will be amplified except target codon to delete.



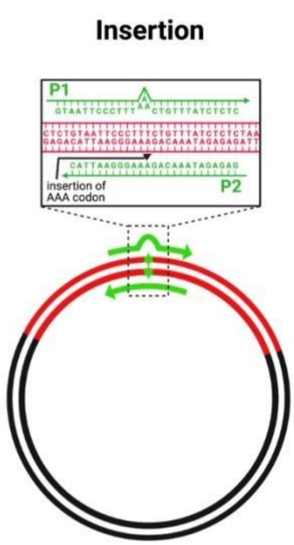

2- **Substitution** (Replacement of either single or multiple a.a residues), using primers with the codon to replace.

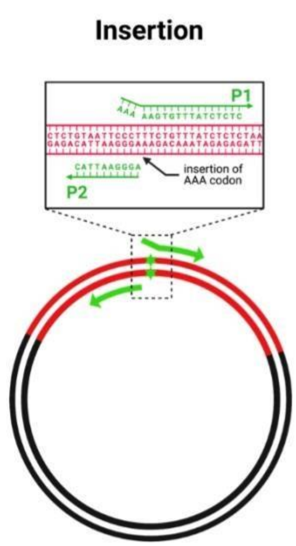
3- **Insertion**, cab be done by two ways:

- Using primer with sequence complementary to the DNA, with the target codon to be inserted in the middle of the primer.

  When primer binds, codon will be inserted and becomes part of the DNA.

- Using primer with target codon at the end of it (like a tail).




Site directed is useful for:

1- Identification of residue (or indeed sequence) is important in the context of some aspect of protein structure and/or function.

Example: for alanine, (alanine scanning) is often used In this context as its side chain is ninbulky, non charged, chemically unstable and is compatible with the fiormation of various structural motifs.

2- Alteration of protein characteristics:

For example: some detergent enzymes have been made oxidation resistance via the replacement of oxidation- sensitive methionine residues on their surface.

As an example of the site directed mutagenesis, consider the replacement of methionine 222 with alanine to stabilize subtilisin for use in laundry detergent.

```
... 218 219 220 221 222 223 224 225 ... amino acid numbering
... AAC GGT ACG TCA ATG GCA TCT CCG ... original DNA sequence
... Asn Gly Thr Ser Met Ala Ser Pro ... original protein encoded

... AAC GGT ACG TCA gcG GCA TCT CCG ... modified DNA sequence
... Asn Gly Thr Ser Ala Ala Ser Pro ... variant protein encoded
```

- OneClick ➐ free online tool that designs and sets up reactions for site directed mutagenesis, it incorporates information of most DNA polymerases and plasmid systems.
- Provides flexibility in choosing the type of mutagenic primers to design, as well as the PCR modes.

Sometimes, improving a protein requires more than one round of mutagenesis which takes months and years of work.

**Site-saturation mutagenesis**

Means: we want to saturate the nucleotide sequence with every possible mutation that will generate every possible amino-acids.It makes all nineteen possible amino-acid substitutions at a particular location. Sometimes we do changes at the DNA level and will get stop codon. When you do the change the amino-acid not change So you can change the codon to introduce a  mutation in a codon ,but that will not change the amino  acid.why? Because there are different codon of same  amino acid and the change will give the same amino acid.

| | Second Base | | | | |
|---|---|---|---|---|---|
| First Base | U | C | A | G | Third Base |
| U | Phe | Ser | Tyr | Cys | U |
| | Phe | Ser | Tyr | Cys | C |
| | Leu | Ser | Stop | Stop | A |
| | Leu | Ser | Stop | Trp | G |
| C | Leu | Pro | His | Arg | U |
| | Leu | Pro | His | Arg | C |
| | Leu | Pro | Gln | Arg | A |
| | Leu | Pro | Gln | Arg | G |
| A | Ile | Thr | Asn | Ser | U |
| | Ile | Thr | Asn | Ser | C |
| | Ile | Thr | Lys | Arg | A |
| | Met | Thr | Lys | Arg | G |
| G | Val | Ala | Asp | Gly | U |
| | Val | Ala | Asp | Gly | C |
| | Val | Ala | Glu | Gly | A |
| | Val | Ala | Glu | Gly | G |

We do random mutagenesis, saturated mutagenesis to  generate every possible amino acid substitute.  The site-saturation mutagenesis will be done by design  degenerate primers that encode multiple amino acids,  these primer contain degenerate codons which are  mixture of codons prepared by DNA synthesis.

-site-directed mutagenesis using degenerate codons  creates a mixture of multiple protein variants.  ..……

Degenerate codons :It means that there is more than  one codon that specifies the single amino acid. Degenerate primer:These are mixtures of primers that are similar, but not identical. These may be  convenient when amplifying the same gene from  different organisms , as the sequences are probably  similar but not identical. This technique is useful  because the genetic code itself is degenerate,  meaning several different codons can code for the  same amino acid. This allows different organisms to  have a significantly different genetic sequence that  code for a highly similar protein. For this reason, degenerate primers are also used when primer  design is based on protein sequence , as the specific  sequence of codons are not known.

…………….

When you want to change the one codon in sequence  These changes or these mutations will done easy by designing a primer or two primers that will introduce the nucleotides.

If I want to change the amino acid to every possible  amino acid (nineteen amino-acid) we do site-saturation  mutagenesis this done by use degenerate primers.  The 103 codon in the sequence encode histidine. We

```
wild type sequence
5'...GATTGCAGCTGCTGTTTTCCACAATTCAGTATTGCCAGACACCGAGC...

primer for site-directed mutagenesis His103Val
   5' GATTGCAGCTGCTGTTTTCGTGAATTCAGTATTGCCAGAC 3'
```

want to change the histidine to another amino acid "valine" this is the site-directed mutagenises(use one primer with one unique sequence) But here

```
wild type sequence
5'...GATTGCAGCTGCTGTTTTCCACAATTCAGTATTGCCAGACACCGAGC...

primer for site-directed mutagenesis His103Val
  5' GATTGCAGCTGCTGTTTTCGTGAATTCAGTATTGCCAGAC 3'

primer mixture for site-saturation mutagenesis His103Xxx
            5' GCTGTTTTCNNKAATTCAGTATTGCCAGAC 3'
```

We want to change the histidine to another possible amino acid this is the site-saturation mutagenesis(use a mixture of primer)

In the primer we add NNK.N: any nucleotides (A,T,C,G).K:G,T). That's mean we have every possible nucleotide at he N and two nucleotides at the K. 4*4*2=32 so this primer mixture consist of 32 primers, so when you do mutagenesis instead of using one primer to change one codon in to a another we use 32 different primers all of them share the same sequence but different in the middle.

We take the gene and put the 32 primer, we will be generating 32 different mutant sequences will change histidine to the other possible 19 amino acid.this done on the DNA level.

| one-letter code | nucleotides |
|---|---|
| N (any) | A, T, G, C |
| B (not A) | T, G, C |
| D (not C) | A, T, G |
| H (not G) | A, T, C |
| V (not T) | A, G, C |
| K (keto) | G, T |
| M (amino) | A, C |
| R (purine) | A, G |
| S (strong) | G, C |
| W (weak) | A, T |

| codon | nucleotide mixture | encoded amino acid |
|---|---|---|
| NDT | AAT | Asn |
| | AGT | Ser |
| | ATT | Ile |
| | TAT | Tyr |
| | TGT | Cys |
| | TTT | Phe |
| | GAT | Asp |
| | GGT | Gly |
| | GTT | Val |
| | CAT | His |
| | CGT | Arg |
| | CTT | Leu |

| codon | nucleotide mixture | encoded amino acid |
|---|---|---|
| VMA | AAA | Lys |
| | ACA | Thr |
| | GAA | Glu |
| | GCA | Ala |
| | CAA | Gln |
| | CCA | Pro |
| ATG | ATG | Met |
| TGG | TGG | Trp |

# Random mutagenesis

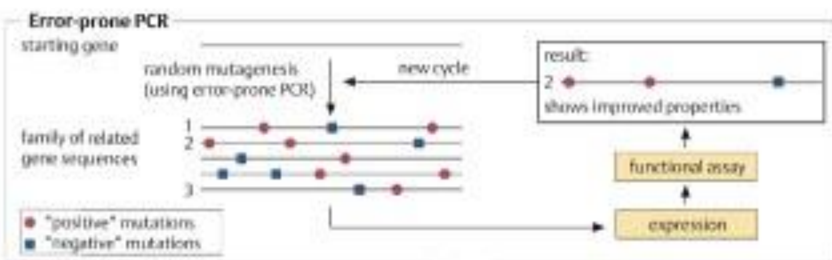
Random mean I don't know the location of the mutations and don't know what will bet this mutation. Random mutagenesis done by using Error-prone PCR. We design 2 primers that will amplify the gene, and no changes in the sequence of primer. Here is the opposite of the regular

PCR, we want the polymerase to make mistakes to introduce mutation that generate amino acid substitution. In error-prone PCR we use a polymerase just like Taq polymerase to lack proofreading activity.

In error-prone PCR reaction conditions are controlled in order to reduce replication fidelity by:

1. The use of DNA polymerase such as Tag polymerase, which lacks proofreading
capability.
2. Replacement of the polymerase's natural cofactor (Mg2+) with Mn2+
3. Including suboptimal relative concentrations of dNTPs in the reaction mixture. Small amount

We change the conditions that have been approved in regular PCR.




In the end of the regular PCR all the products it’s identical but in error-prone PCR the products have a different sequence, so we get a library of PCR products.

We have starting gene —> random mutagenesis by error prone PCR to introduce mutations—> library of sequences—> express them and do functional assay. Functional assay means you want to express mutant sequences to produce mutant proteins and then you have to test these mutant proteins to select mutant protein that will improved.

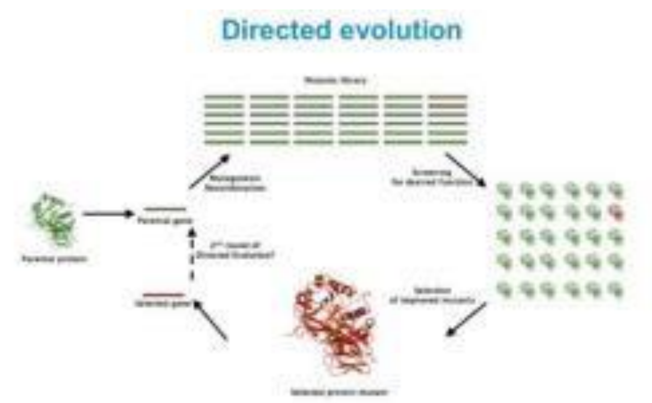
## Directed evolution

• The introduction of random mutations into the gene coding for the protein of interest, thus generating a large library of gene variants.This process is known as **diversification**.

The library of variants is expressed and screened to identify variants displaying enhanced target characteristics.

The gene coding for the desired variant is then cloned and sequenced in order to identify the exact amino acid alterations present.

Diversification is usually achieved in practice using a technique termed error- prone PC or gene shuffling.

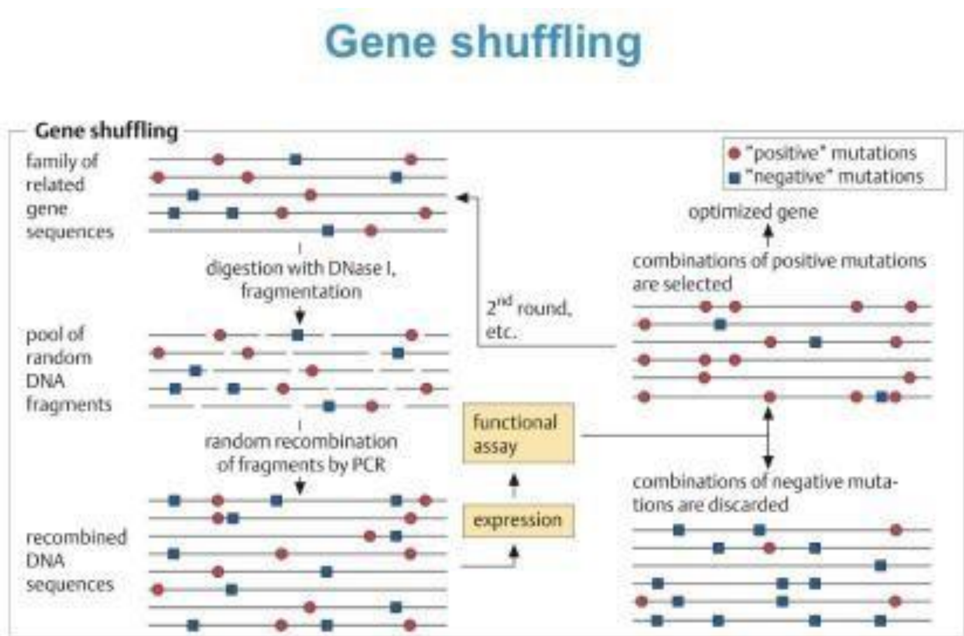



## Gene shuffling

Gene shuffling facilitate the recombination of several individual beneficial mutations from different mutants into a single protein.

The process entails fragmentation of several mutant genes, generally into lengths in the region of 50 base pairs.

The fragment mix is then subject to PCR, but in the absence of a primer. Overlapping fragments will anneal with each other, allowing eventual extension to full gene sequence size. The recombined genes can be cloned, expressed and screened for improved target characteristics.




You take the same gene from different organisms and you cut them with DNAs1 or do error-prone PCR lead to have a library of mutant gene then cut in DNAs1. Now we have millions of fragments coming from thousands of mutant gene and do random combination using PCR. So the mutation could end up of the mutation from the other gene. Some of these genes will have the positive impact of the protein properties and some of gene have the negative, we take the once of the positive impact and try to do the cycle again to reach to a point that have a very improved protein.

<u>Combined different mutations from different mutant genes ➐ **gene shuffling**</u>

## Protein post-translational modification

- Polypeptides undergo covalent modification, either during or after their ribosomal synthesis is a process called post-translational modification, PMT).
- PTMs are characteristic particularly of eukaryotic proteins and are introduced by specific enzymes or enzyme pathways.
- Many occur at the site of a specific characteristic protein sequence (signature sequence) within the protein backbone

Post-translation modification: addition of chemical group or the removal of amino acids or other processes to make the protein functional

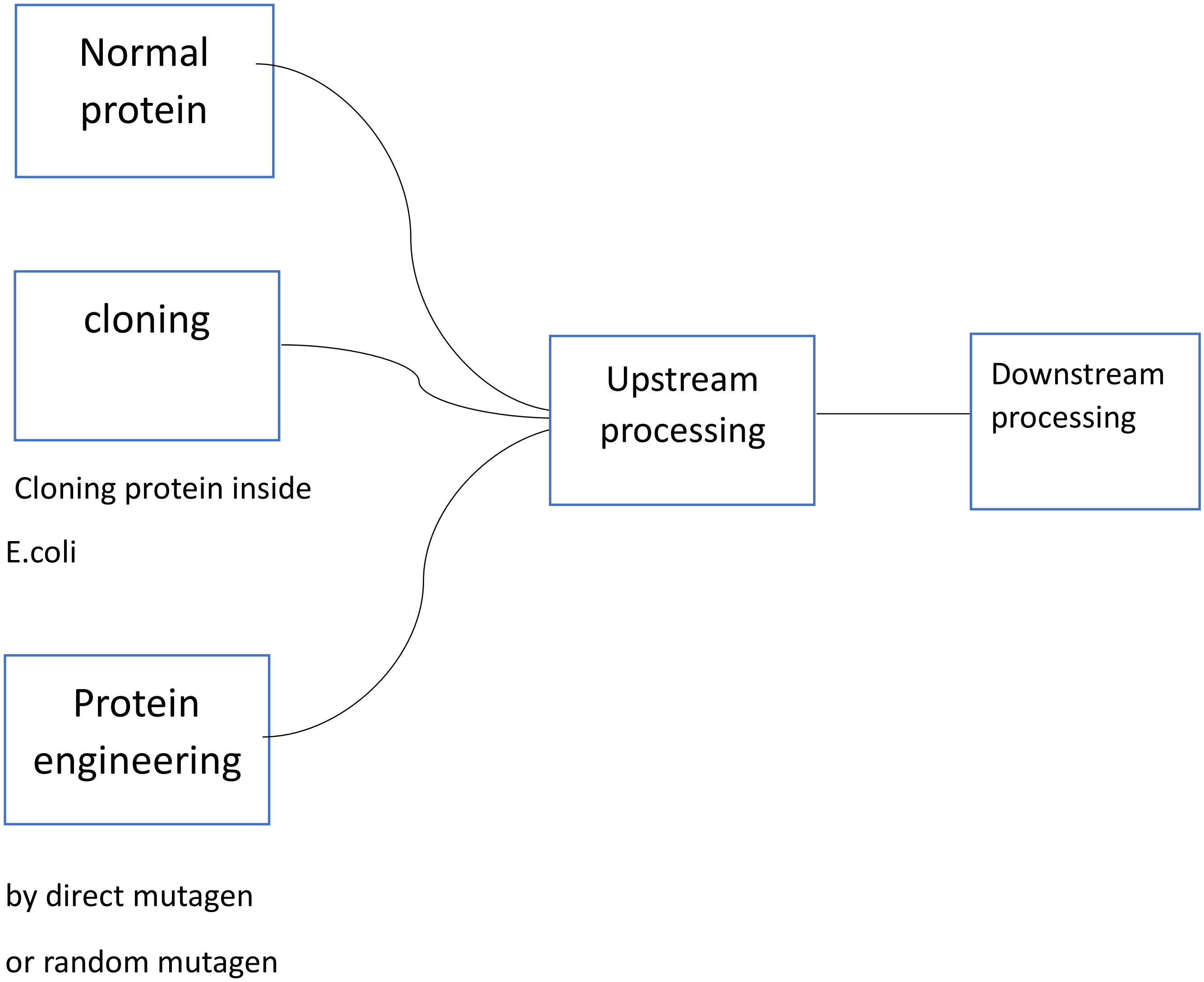
# Protein post-translational modification

**Table 2.7** The more common forms of post-translational modifications that polypeptides may undergo. Refer to text for additional details.

| Modification | Comment |
|---|---|
| Glycosylation | For some proteins glycosylation can increase solubility, influence biological half-life and/or biological activity |
| Proteolytic processing | Various proteins become biologically active only on their proteolytic cleavage (e.g. some blood factors) |
| Phosphorylation | Influences/regulates biological activity of various regulatory proteins including polypeptide hormones |
| Acetylation | Modulation of target protein activity |
| Acylation | May help some polypeptides interact with/anchor in biological membranes |
| Amidation | Influences biological activity/stability of some polypeptides |
| Sulfation | Influences biological activity of some neuropeptides and the proteolytic processing of some polypeptides |
| Hydroxylation | Important to the structural assembly of certain proteins |
| γ-Carboxyglutamate formation | Important in allowing some blood proteins to bind calcium |
| ADP-ribosylation | Regulates biological activity of various proteins |
| Disulfide bond formation | Helps stabilize conformation of some proteins |

Now let’s say we want to express a gene to produce a protein now to do that we can start with different thing to produce the desired protein so this protein could be a normal protein, or it could be engineered this depend on what is the protein that I want to produce

## Protein Production




– Production of proteins is a long and painstaking process

– Two major phases used in producing proteins

- Upstream processing includes the actual expression of the protein in the cell
- Downstream processing involves purification of the protein and verification of the function; a stable means of preserving the protein is also required

Now whether I do cloning, natural production, protein engineering to produce protein I have to do upstream processing because the upstream involve the actual expression of protein Note:

I can give the bacteria special food to enhance the production of protein

## Protein Expression: Upstream Processing

Now as we said that upstream is the actual expression so we need a host cell this host cell could be plant, animal cell, fungi, bacteria, or it could be a whole animal depending on the protein that I want to produce

Lets talk about bacteria as a host cell, we will have some advantage such as

1- Easy to grow in large number
2- Divide quickly
3- Haploid so it will be easily genetically engineered

– Selecting the cell to be used as a protein source

• Bacteria
– Fermentation processes are well understood
– Cultured in large quantities in short period of time
– Relatively easy to alter genetically
– Can increase level of production of a bacterial protein by introducing additional copies of relevant gene to the host cell via recombinant DNA technology How is the gene is under control of expression?
–Most common bacterial species used is E. coli

Now we have to know something in this course we are talking about biotechnology as industry so its true that I can produce normal amount or protein for research for example but just like any industry in the world the main goal is to make profit(money) so to make more money I have to produce the maximum amount of protein inside each bacterial cell and to do that we do something called overexpression

You have to know that the overexpression is not a good thing because the bacteria will be under metabolic pressure (most of the energy will be used to produce the protein)

Now what will result from overexpression is aggregating of the protein to form inclusion bodies that because of the high concentrations of protein

The inclusion bodies could include a fully folded protein or partially folded or not folded protein

Of course, in some way you can take the protein you produced from these inclusion bodies

Reminder:
What control the level of expression is the promoter

Now there is an important step (it's not essential but it will help us a lot especially in the downstream process) and it is making fusion protein

Fusion protein is adding amino acid to the protein I want to produce by adding tag to the protein coding gene
Now why did we say it is important although it's not essential?

Because the fusion protein will make the downstream process easier and you have to know that the downstream process is the most expensive process and we just said that the main goal in the industry is to make money so by the fusion protein we can reduce the cost so we will make more money

And sometimes to get a fully folded protein we add the tag because when we do expression to a human inside a bacterial cell we will not get always a fully folded protein because we are expressing it in another cell than the normal cell

So the advantages for the fusion protein are:

1- Increase the solubility (can help to prevent the producing of inclusion bodies)
2- To make the protein fold correctly
3- For purification
4- To release the protein outside the cell

Note:

Intracellular❼ inside the cell
Extracellular❼ outside the cell

Now if I do expression in bacterial cell how can I secrete the protein outside the cell?

The tag will work as a signal that will allow the cell to know that this protein must be secreted

Protein Expression: Upstream Processing

– Selecting the cell to be used as a protein source

• Bacteria- majority of proteins synthesized naturally by E coli are intracellular – Most cases resultant foreign protein accumulates in cell's cytoplasm in form of insoluble clumps = inclusion bodies which must be purified from other proteins

• Sometimes the genetically engineered E coli produce the desired protein in the form of a fusion protein
– Target protein is fused to bacterial protein so an additional step is necessary to break the two apart

» Fused bacterial protein is usually an enzyme that will bind to its substrate and can be attached to a purification column

Pic

note

Now you have to know that about 50% of human protein are post modified and we already know that bacterial cells don't carry post translation modified

– There are pros and cons to expressing proteins in bacteria

• Prokaryotic cells are unable to carry out processes such as glycosylation

One of the disadvantages of the expression in bacteria is that the protein may result unfunctional

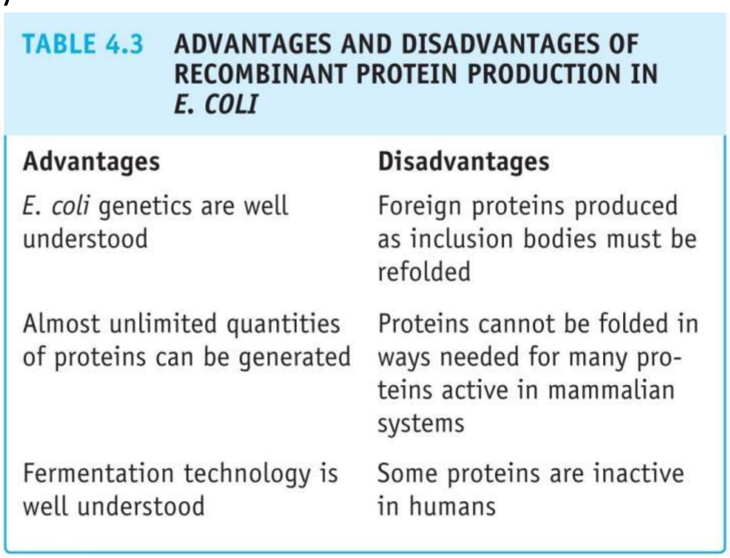


**TABLE 4.3 ADVANTAGES AND DISADVANTAGES OF RECOMBINANT PROTEIN PRODUCTION IN *E. COLI***

| Advantages | Disadvantages |
|---|---|
| *E. coli* genetics are well understood | Foreign proteins produced as inclusion bodies must be refolded |
| Almost unlimited quantities of proteins can be generated | Proteins cannot be folded in ways needed for many proteins active in mammalian systems |
| Fermentation technology is well understood | Some proteins are inactive in humans |

Genetically engineered bacteria can be grown in large scale fermenters (anaerobic, without oxygen) or bioreactors (aerobic, with oxygen)

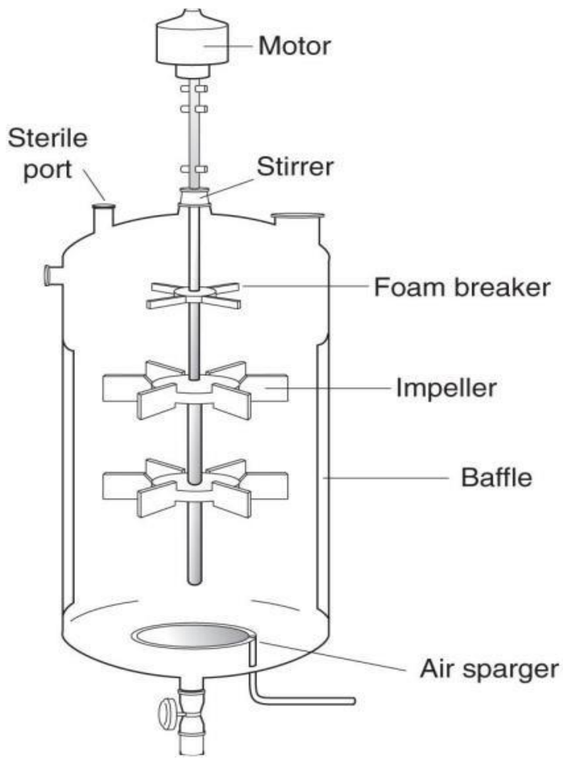
• Computers monitor air in bioreactors keeping oxygen levels and temperature ideal for cell growth

• When phase of cell growth is highest- the bacteria promoter must be activated to stimulate foreign gene expression

– Must activate gene in recombinant organism after the organism has completed synthesizing important natural proteins needed for its metabolism

Fermenter➐ large tank to grow bacteria without oxygen (anaerobic)

Bioreactor➐ large tank to grow bacteria with oxygen (aerobic)




Typical fermenter will contain liquid media and inside the fermenter will be the bacteria

Impeller: there job is to mix everything inside together the purpose of mixing is to make sure that bacteria have enough food

Foam breaker: to prevent the forming of foam, this foam is very bad for protein

Baffle: will make the surface of fermenter not smooth to prevent the sticking of bacteria on the surface

Air sparger: this device provides for bacteria like air bubbles(contain oxygen)

Sterile port: connect inside them sensor to examine the temperature, ph, concentration of molecule and these sensor will be connected to computer software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